PRANAV DUBEY

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Education

Vellore Institute of Technology, Bhopal 2023 – 2027
B.Tech in Computer Science Engineering *CGPA: 8.07*

Technical Skills

- **Languages & Frameworks**: Python, FastAPI, Streamlit
- **Generative AI & LLMs**: LLM Integration, Context Engineering, RAG, LangChain, ChromaDB, Ollama, Hugging Face
- **Machine Learning & Deep Learning**: PyTorch, TensorFlow, Keras, Scikit-learn, CNNs, LSTMs, YOLOv8, NLP
- **Data & Analytics**: NumPy, Pandas, Matplotlib, SQL, PostgreSQL
- **MLOps & Tools**: AWS, Docker, Git, REST APIs, Postman

Research Experience

AQUANET: YOLOv8-based Detection of Objects Displaced During Floods *Co-author — Under Journal Review*

- Co-authored a YOLOv8-based computer vision model with Prof. Brejesh Lall (IIT Delhi) and PhD scholar Gurjant Singh for detecting high-value objects submerged in post-flood water and mudflow, achieving 93.0% mAP50.
- Benchmarked YOLOv8 s/m/l variants by reproducing training runs and validating accuracy, precision, recall, and mAP, confirming reproducibility of the reported results.
- Prepared the training dataset, ran ablation studies, and co-wrote the manuscript now under journal review, collaborating across the full research lifecycle from data to submission.

Projects

Congenital Heart Defect Prediction System | *Spring Boot, FastAPI, PyTorch, Docker* **Demo** | **GitHub**

- Trained a dual-branch ConvNeXt for chest X-ray classification (Normal, ASD, VSD) using CLAHE, feature fusion, and focal loss, achieving 82.6% accuracy and 0.83 macro F1.
- Designed and shipped a FastAPI + Spring Boot backend exposing 22+ REST APIs on PostgreSQL and Docker, serving predictions and patient history end-to-end.
- Secured medical data access end-to-end with JWT authentication and role-based access control (RBAC), restricting actions by clinician/admin role.

AI Video Surveillance: Anomaly Detection System | *PyTorch, YOLOv8, FastAPI, Docker* **Demo** | **GitHub**

- Engineered an end-to-end CCTV anomaly detection system (ResNet50 + LSTM) trained on 1.37M+ frames from 1,900 UCF-Crime videos.
- Outperformed the Sultani et al. (2018) baseline by 6.5 points, reaching 0.803 AUC-ROC (vs. 0.754), with YOLOv8 and Grad-CAM for visual explainability.
- Deployed the model behind a FastAPI backend supporting asynchronous video upload and inference, live on a Hugging Face Spaces demo.

Cross-Source News Verification | *Python, LangChain, ChromaDB, Ollama, FastAPI, React* **GitHub**

- Designed a RAG pipeline that retrieves related news coverage and flags where independent sources agree, conflict, or offer no supporting evidence, with grounded citations for every claim.
- Improved retrieval precision by removing duplicate syndicated articles and tuning similarity thresholds to suppress low-relevance matches.
- Implemented local inference with Ollama and ChromaDB, built modular LCEL components, wrote 225 offline tests, and shipped the app via FastAPI and React.

Achievements

- Scored 97% in the NPTEL Marketing Analytics certification conducted by IIT Kharagpur, ranking among the top 5% of certified learners.
- Participated in the Shell.ai Hackathon 2025, developing and optimizing machine learning models for sustainable aviation fuel blend property prediction in a national-level industry challenge.